
COM 480 - Data Visualization

MILESTONE 3



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

AquaViz

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1 Project idea

Water is one of the most essential resources on which human life and economic activity depend. Yet climate change, falling rainfall and shrinking glacial reserves are turning water scarcity into a critical issue across large parts of the world. In addition, water scarcity is often considered as a single global issue, but the underlying causes and consumption patterns differ across countries. For these reasons, we decided to design a data visualization that allows users to compare countries, track changes over time, and understand the water consumption patterns of each country.

AquaViz is an interactive dashboard that lets a general audience explore how patterns of consumption *and* sectoral allocation (agriculture, industry, households) relate to scarcity levels over time. We worked with the [Kaggle Global Water Consumption Dataset \(2000–2024\)](#). It contains 500 observations covering 20 countries across 25 years and 10 variables: total consumption (bn m³/year), per-capita use, the share going to agriculture, industry and households, rainfall impact, groundwater depletion, and a categorical scarcity level (Low / Moderate / High). The data was already clean, so we only needed light preprocessing (mainly grouping by country and year for the time-series view).

1.1 Main idea

Designed for a general audience interested in environmental issues, the visualization aims to make water scarcity more understandable by showing how water use differs across countries and sectors, and by emphasizing global disparities in access and consumption. Since the dataset combines information about countries, sectoral allocation, time, and scarcity levels, it allowed us to investigate how patterns of water use are associated with different levels of water scarcity across countries and years. Based on this, we formulated the following main question:

How does water consumption and its distribution across sectors contribute to water scarcity levels across countries over time?

The plan was the following: an interactive world map where each country is coloured by its level of water scarcity, and a side panel that updates whenever the user clicks on a country to show the more detailed numbers (a doughnut chart for the sectoral breakdown, a bar chart for per-capita consumption over the years, and a few key indicators on top). A year slider at the top of the page ties everything together, so the whole dashboard moves in time as the user drags it.

2 From Initial Sketch to Final Design

2.1 The Milestone 2 sketch

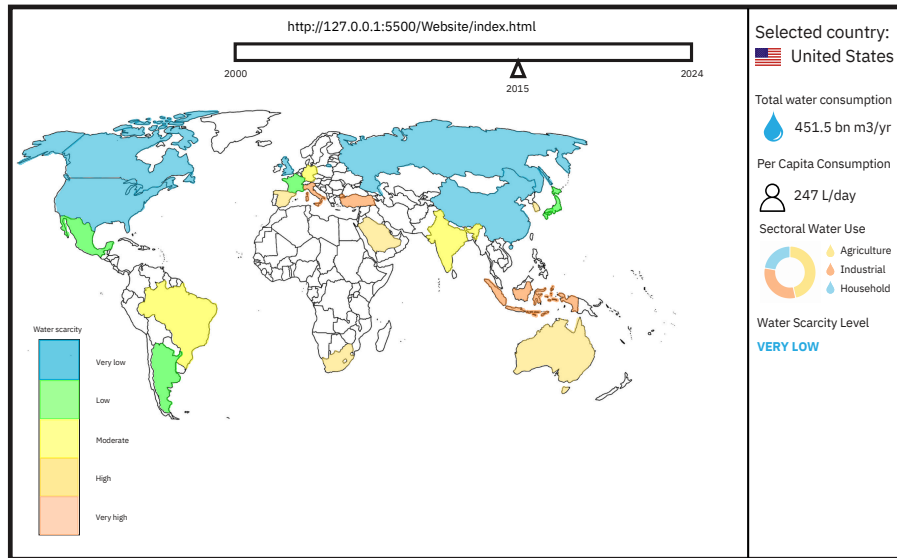


Figure 1: Sketch of the Website [1]

2.2 The final design

The final dashboard collapses every view into one synchronised, two-column layout, driven by a global year slider in the header.

2.3 Design decisions

A few choices made between Milestone 2 and the final version (While some parts were implemented as originally planned, others were adjusted during the development process):

- **One unified dashboard.** Each view shares the same selected year and the same selected country. We place a fixed dashboard on the right side of the website to improve readability and maintain consistency.
- **A global year slider in the header.** Time is not just a filter, it's the main axis of the story. Putting the slider at the top makes it obvious that it controls everything below. We also added a feature that allows users to view changes over time as an animation by clicking the play button next to the time bar after Milestone 2.
- **A doughnut chart rather than a stacked bar** for the sectoral split. With only three categories (agriculture, industry, households), the doughnut reads faster.
- **A bar chart rather than a line chart** for per-capita consumption. We wanted users to perceive each

year as a discrete unit (and to be able to highlight “the year the slider is currently on” and the historical fact), which is easier with bars than with a continuous line.

- **Comparison between two countries.** We added a feature that allows users to compare two countries. By holding the Shift key and selecting two countries, users can compare their information side by side in the dashboard.
- **Intro page with a water drop illustration.** We added a intro page with a water drop illustration, where the user can simply click the water drop and start using the visualization.
- **A single turquoise sequential scale for the choropleth.**

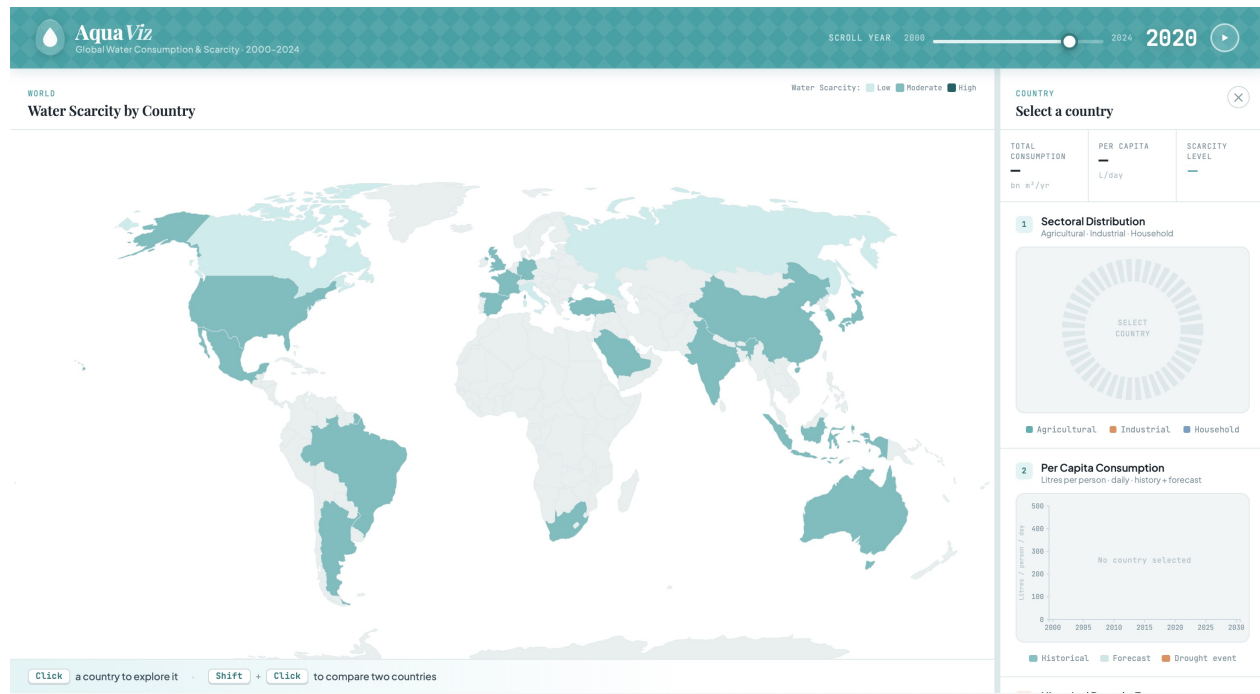


Figure 2: Sketch of the Website AquaViz

3 Visualisations & Interactive Story

The intended path through the dashboard is straightforward: drag the year slider (or hit the play button), scan the map to spot regions under stress, click a country to drill down, and read the more detailed indicators on the right. From there, the user can also compare two countries side by side or look at the historical events that shaped the situation.

3.1 Choropleth world map

Built on Natural Earth, using D3 v7, `topojson-client` and `world-atlas@2`. Each country is filled according to its **Water Scarcity Level**. Hovering shows a tooltip with the country name, the scarcity level and the

total consumption in $\text{bn m}^3/\text{year}$. Clicking selects the country and zooms in on it. The user can pan and zoom freely with `d3.zoom`, and a double-click on the ocean resets the view to its starting position.

3.2 Indicator strip

Three indicators are recomputed every time the user changes the selection: the total consumption ($\text{bn m}^3/\text{year}$), the per-capita use (litres per day), and the textual scarcity level. They act as the bridge between the map (which is global) and the more detailed charts below (which are country-specific).

3.3 Sectoral doughnut

Three arcs, one per sector (agricultural, industrial, household). The slice size is the share of that sector in the selected country, and the centre of the doughnut shows the country's total consumption for the selected year.

3.4 Per-capita bar chart (2000–2024)

Litres per person per day, year by year, for the country currently selected. The bar matching the year on the slider is drawn fully opaque and overlaid with a dashed vertical line, so the user can read both the long-term trajectory and the year they're currently looking at in one glance.

3.5 Forecast (2025–2030)

On the same bar chart, we also show a short forecast for the years 2025 to 2030, drawn as lighter turquoise bars with a dashed outline so they read clearly as predictions rather than real data. The values come from a simple damped linear trend fitted on the last ten years, where the slope is clipped to avoid runaway predictions and is shrunk a bit each future year so the curve gradually flattens out.

3.6 Historical drought events

Beneath the bar chart, the side panel shows a small list of major water-related events for the selected country, when our research turned up any: 2003 heatwave in France, Cape Town's Day Zero in 2018, and so on. Each event is a small card with the year, a short description and the source. Clicking a card jumps the year slider to that year, which makes it easy to see what the country's numbers looked like at the time of the crisis. The same events also appear as orange pins on the bar chart itself, so the timeline reads at a glance.

3.7 Comparing two countries

The dashboard works around a single selected country by default, but **Shift + Click** on another country switches the side panel into a comparison view. Both countries get a colour tag (turquoise and orange), and the same indicators we showed for a single country are now shown side by side: total consumption, per-capita use, and scarcity level, with a small delta badge in the middle that says by how much one is higher than the other in percent. The two sectoral doughnuts sit next to each other so the user can immediately tell, for instance, that a country leans more on agriculture than the other. The per-capita bar chart turns into a grouped bar chart with one bar per country per year. A swap button flips A and B in a click, and any further **Shift + Click** on the map replaces the second country.

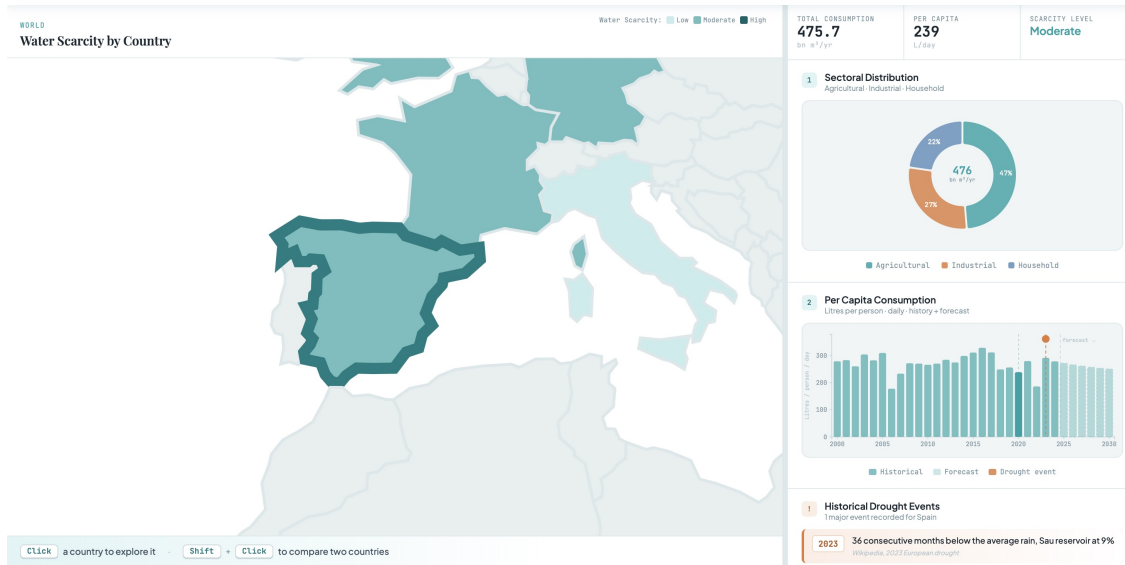


Figure 3: Dashboard after a country selection.

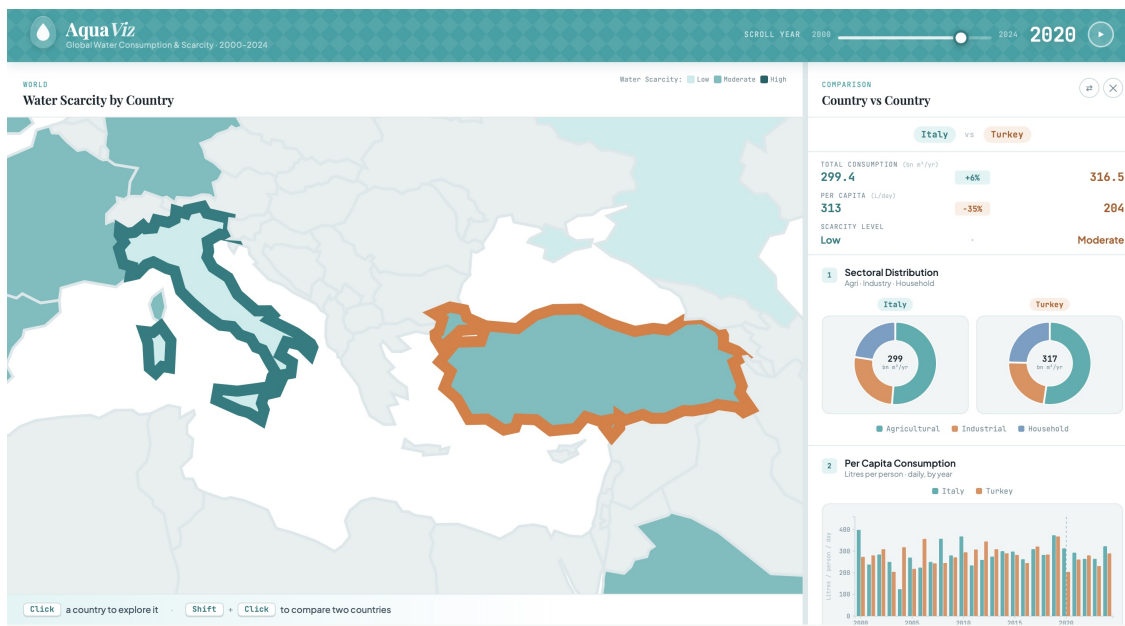


Figure 4: Comparison between two country.

4 Technical Implementation

Stack. Vanilla HTML/CSS/JavaScript, D3 v7, topojson-client 3, world-atlas 110m, and three Google Fonts (*Playfair Display* for headings, *Plus Jakarta Sans* for the body, *JetBrains Mono* for numbers and labels). No build step, no framework: the whole site is three files and loads in a single request.

Code organisation. `script.js` is split into numbered sections (global state, helpers, status bar, slider, map, zoom helpers, tooltip, charts, central update, data loading, bootstrap). Every view follows the same pattern: an `init` function builds an empty SVG skeleton, then a `render` function binds the real data on top. That way the layout doesn't "jump" when the dataset finishes loading.

Geo ↔ CSV reconciliation. The TopoJSON file we use doesn't always populate `properties.name`, so we had to attach country names ourselves through a `NAME_BY_ID` table covering 90+ ISO codes. On top of that, a second table (`GEO_NAME_TO_CSV`) handles the cases where the geographic name and the name in our CSV simply don't match (*United States of America* → *USA*, *United Kingdom* → *UK*, *Republic of Korea* → *South Korea*, *Russian Federation* → *Russia*, etc.).

Transitions and responsiveness. All SVGs use `viewBox` and `preserveAspectRatio`, so the dashboard scales down to a tablet layout without breaking. CSS media queries push the side panel below the map when the viewport drops below 680 px.

Accessibility. Every chart has a `role="img"` and an `aria-label`, the slider has explicit `min`, `max` and `step`, focus rings are visible, and the palette was checked against WCAG AA for text on background.

Resizable side panel. A small drag handle sits between the map and the side panel. The user can pull it left or right to give either side more room, and their preferred width is saved in `localStorage` so the layout sticks across reloads.

4.1 Challenges

The biggest constraint is that the dataset covers only 20 countries, so the choropleth cannot be globally complete. Instead of hiding the rest of the world, we draw all countries on the map and grey out those without data, with a tooltip that says so explicitly. We then put most of our effort into making the experience for the 20 covered countries as rich as possible, with the comparison, the forecast, the per-country events, and the play button on the time bar from 2000 to 2024.

We had to match country names between the GeoJSON and the CSV. The TopoJSON we use does not always expose country names, and several countries appear under different labels in the two sources (*United States of America* vs. *USA*, *Republic of Korea* vs. *South Korea*, etc.). We solved this with two small lookup tables: one from numeric ISO codes to country names, and one from GeoJSON names to CSV names.

5 Tools and Methods

- **Tools and deployment:**

- **HTML5, CSS, JavaScript (ES6)** for the DOM skeleton, the layout and the application logic.
- **D3.js** for the interactive choropleth, the doughnut chart, the bar chart, and the zoom/pan behaviour.

- **GeoJSON / TopoJSON** (`world-atlas@2`) for the country polygons, decoded on the fly with `topojson-client`.
- **GitHub Pages** for deployment. Since the project is fully static (three files plus the dataset), we didn't need a server: the site is served straight from the repository.
- **Course concepts we relied on:**
 - **Maps.** The main view is a choropleth, built with GeoJSON and the D3 Geo module to project and render the polygons.
 - **Perception.** Country polygons are the marks; colour (a sequential turquoise ramp) is the channel for scarcity. The sectoral colours are categorical, deliberately picked to be distinct from the choropleth ramp so the two scales never get confused.
 - **Interactions.** The year slider and the map clicks update every linked view at once, which keeps the cognitive load low: the user only has two things to manipulate but gets four views in return.
 - **Storytelling.** The side panel is meant to be read top to bottom (KPIs, then sectoral breakdown, then long-term per-capita trend), so it tells a small per-country story once a country is selected.

6 Implementation steps

We broke the project down into three phases:

- **Phase 1 — MVP base.** Clean the dataset, set up the mapping between the CSV country names and the GeoJSON features, and write the basic HTML/CSS skeleton.
- **Phase 2 — Core visualisation.** Implement the choropleth, the year slider and the linked side panel (KPIs + doughnut + bar chart). At the end of this phase the user could pick a year, click on a country and see the main indicators.
- **Phase 3 — Extensions, polishing and deployment.** Add the play button on the time bar, the 2025–2030 forecast on the per-capita chart, the per-country historical events, the two-country comparison view, the intro page with the water drop, and the drag-to-resize side panel. Refine the visual style (palette, typography, spacing), add zoom/pan on the map and the responsive behaviour, then deploy the static version to GitHub Pages.

7 Peer Assessment — Contributions

Member	Main contributions
Mohammad Massi RASHIDI	D3 implementation (map, zoom, tooltip, doughnut, bar chart), UI/UX design, Milestone 1 EDA, Milestone 2 website skeleton, Process book
Anna ISTEPANYAN	Implementation of the extension of Milestone 2 on the website, Milestone 1 Dataset and Problematic, Milestone 2 Project Idea
Yoojin OH	Website ideation, Screencast planning and producing(demo & insights), Milestone 1 Related Work research, Milestone 2 Overall integration

Here is the link of the skeleton website : [Website AquaViz](#)

References

- [1] World Map With Countries. *5 Free Printable Blank World Map with Countries Outline in PDF*. Accessed: 2026-05-24. URL: <https://www.bing.com/images/search?view=detailV2&ccid=pV%2bMsvY6&id=6341B0573F30DFCEB0199A14E5ACE40DAACF2111&thid=OIP.pV-MsvY67wHZc96olykFagHaEJ&mediaurl=https%3a%2f%2fworldmapwithcountries.net%2fwp-content%2fuploads%2f2018%2f07%2fblank-map-of-the-world-1.gif&cdnurl=https%3a%2f%2fth.bing.com%2fth%2fid%2fR.a55f8cb2f63aef01d973dea89729056a%3frik%3dESHpqg3krOUUmg%26pid%3dImgRaw%26r%3d0&exph=1370&expw=2446&q=map+world+blank&FORM=IRPRST&ck=4ADC363E221741C4C5E6A2E2F0F63234&selectedIndex=0&itb=0&ajaxhist=0&ajaxserp=0>.